

M5SHARK

OWNER GUIDE

Rewritten Reference Edition

Prepared under the name Domantas Uba

Setup, device reference, firmware paths, safety notes, troubleshooting, and links for the M5Shark hardware ecosystem.

Exact device-family guidance

Lab-first and receive-first safety

Firmware and accessory reference

FAQ, troubleshooting, and legal reminders

For lawful education, development, repair, interoperability study, and authorized testing only.

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1. How to Use This Guide

This guide is a Domantas Uba rewritten owner reference for the M5Shark hardware ecosystem. It keeps the original intent: help owners identify the device family, choose a correct firmware path, handle accessories safely, and troubleshoot common setup problems.

Use it as a practical checklist before firmware updates, RF experiments, NFC/RFID work, IR learning, SDR exploration, GPIO wiring, or accessory testing.

Core rules: identify the exact model, start with receive-first learning, work only with owned or authorized targets, and stop when the device behaves unexpectedly.

Coverage: device family overview, first-use flow, firmware routes, Bruce-style setup, Universal RF safety framing, links, FAQ, and legal/safety reminders.

2. Product Family Overview

M5Shark / SHARK RF: portable RF learning and Flipper-style education. Confirm the hardware revision, charge the battery, add a known-good microSD card when needed, and use firmware meant for the exact revision.

ESP32 / Bruce / Marauder devices: open-source ESP32-based lab tools for Wi-Fi, BLE, IR, and RF education. Select the exact board target before flashing and keep tests on owned networks or lab devices.

Chameleon Ultra V2: RFID/NFC study platform for authorized tag and card training. Test only cards, tags, and systems you own or have permission to assess.

H4 + R10 SDR: receive-first spectrum learning device. Begin with observation, choose the right antenna, and follow local radio-frequency rules.

Universal RF Research Tool: compliance-review and shielded-lab reference platform. Treat it as controlled-lab equipment, not as a public-space operating tool.

3. From Box to First Safe Use

1. Identify: write down product family, board revision, included accessories, and the current About/Firmware screen.
2. Inspect: check USB-C port, cable, enclosure, display, buttons, battery area, antenna connector, and microSD card.
3. Set up: use only the firmware route that matches the device family. Similar-looking hardware can still need different firmware.
4. Learn: begin with receive-first, demo, scan, log, or owned-device workflows before advanced modes.
5. Troubleshoot: if something fails, check exact model, firmware screen, cable, SD card, power state, and accessory compatibility before changing firmware again.

4. Device Chooser

M5Shark Device / SHARK RF Research Device: portable RF multi-tool for RF, NFC, RFID, IR, GPIO, authentication demonstrations, hardware learning, and authorized security education.

ESP32-DIV V2.0 Development Board: ESP32-S3 handheld toolkit for Wi-Fi/BLE learning, 2.4 GHz research, Sub-GHz study, IR labs, SD logging, and touchscreen demos.

RF CLOWN V2.0: ESP32/nRF24 RF learning board for 2.4 GHz protocol study, portable RF experiments, firmware study, OLED menu demos, and classroom RF concepts.

nRFBBox-v3: ESP32 wireless research tool for 2.4 GHz observation, BLE discovery, channel scanning, protocol study, and open-source firmware learning.

Mini ESP32 Marauder and ESP32 Marauder: devices for Wi-Fi/BLE education, wireless diagnostics, CTF labs, lab scanning, educational demos, and firmware exploration.

ESP32-C5 Marauder / Flipper expansion: expansion or standalone module for Flipper expansion workflows, ESP wireless learning, GPS-assisted demos, large-screen viewing, and authorized 433 MHz lab tests.

Chameleon Ultra V2: RFID/NFC reader, writer, and emulator family for authorized card technology study and lab tag testing.

T-Embed CC1101 Plus / TDeck Plus: ESP32-S3 handheld family for larger-screen RF learning, keyboard-heavy workflows, Sub-GHz study, and portable protocol labs.

NM-RF-Hat for Bruce CYD: expansion hat that adds CC1101, nRF24, and PN532-style learning modules to compatible CYD/Bruce setups.

H4 + R10 SDR Device: SDR and portable spectrum platform for receive-first signal education and firmware exploration.

Universal RF Research Tool / Universal Signal RF platforms: multi-band lab platforms for bench learning, shielded-lab review, antenna planning, and RF safety education.

5. Device Family Details

M5Shark / SHARK RF: STM32-class Flipper-style learning device with 315/433 MHz RF, SMA antenna port, monochrome display, USB-C, microSD support, Li-Po battery, and menus for RF, IR, NFC, RFID, GPIO, files, and help. Use only firmware confirmed for the exact hardware revision.

ESP32-DIV V2.0: ESP32-S3 board with 16 MB flash, touchscreen, SD slot, USB-C, multiple nRF24 modules, CC1101, RGB LEDs, buzzer, and battery support. Keep transmit-capable work lab-only.

RF CLOWN V2.0: ESP32-WROOM-32U with triple nRF24 layout, OLED display, three-button control, status LED, and IPEX antenna connectors. Favor receive-first workflows and avoid interference framing.

nRFBBox-v3: compact ESP32 WROOM32U platform with nRF24 modules, OLED interface, and onboard controls for channel observation, BLE discovery, and protocol education.

Mini ESP32 Marauder: ESP32-based Marauder device for Wi-Fi/BLE education and diagnostics. Controls vary by build; use only owned or authorized networks/devices.

ESP32 Marauder: Marauder firmware ecosystem device for wireless security training and lab scanning. Keep guidance educational, authorized, and model-specific.

ESP32-C5 Marauder / Flipper expansion: 4-in-1 concept with ESP module, RF, GPS, display, external antennas, and rechargeable battery depending on version. Confirm compatibility before connecting it to another device.

Chameleon Ultra V2: dual-frequency RFID/NFC training ecosystem with USB-C/Bluetooth control depending on build. Use only with authorized tags, cards, and systems.

T-Embed CC1101 Plus / TDeck Plus: ESP32-S3 handhelds with display and RF options. Match firmware to the exact board and antennas to the correct region/frequency band.

NM-RF-Hat for Bruce CYD: RF/NFC expansion hat. Verify pinout, voltage, accessory support, and firmware target before powering the hat.

H4 + R10 SDR: two-in-one SDR family with display, controls, USB-C, or host features depending on model. Keep it receive-first unless you have explicit legal authority.

Universal RF Research Tool: aluminum handheld lab reference with cooling, battery, switches, indicators, charging input, and antenna/channel bank. Use only for legal compliance review in controlled environments.

6. Modules, Accessories, and Shared UI

CC1101 Wireless Module: Sub-GHz module. Use 3.3 V logic, a correct antenna, and legal bands only.

NRF24L01+PA+LNA Module: 2.4 GHz module. Use stable 3.3 V power and a 2.4 GHz matched antenna.

2.4 GHz Wi-Fi/Bluetooth Antenna: match SMA/RP-SMA connector type and frequency before use.

Micro SD Card Module / Storage Board: use for logs, configuration files, profiles, and safe backups.

PCB Protoboard: use only for low-voltage electronics practice and careful bench wiring.

A consistent owner interface should show battery, storage, accessory status, current mode, safe-start action, files/log shortcut, settings, and help. Error messages should be direct: antenna missing, storage not ready, low battery, wrong accessory, or wrong firmware.

7. Firmware and Setup Paths

Wrong firmware can break display, buttons, storage, radio modules, boot behavior, or menus. Always match the exact device family before flashing.

M5Shark / SHARK: use firmware supplied for the SHARK RF device or a Flipper-style build confirmed for the exact M5Shark hardware revision.

Bruce-compatible products: use the Bruce website, Bruce web flasher, Bruce GitHub, or M5Burner only where a matching build exists. Bruce is not universal firmware for every STM32/SHARK device.

ESP32-DIV V2.0, RF CLOWN V2.0, nRFBBox-v3, ESP32 Marauder, Chameleon Ultra V2, and H4 + R10 SDR each need their own project resources and exact board/revision target.

Universal RF Research Tool: do not assume generic firmware. Treat manuals as safety/hardware references and verify exact model, charger, compliance status, and legal context.

8. M5Shark Flipper-Style Firmware Guide

The M5Shark Device is handled here as a Flipper-style RF learning device. Use only firmware confirmed for its exact hardware revision.

Official Flipper Zero firmware is the safest reference for stable update workflows, recovery concepts, qFlipper behavior, and release-channel expectations. Normal owners should prefer Release over RC or Dev channels.

Flipper Lab can help with compatible browser updates, apps, and file management. qFlipper is useful for desktop backup, restore, update, and recovery workflows.

Custom firmware such as Unleashed, Momentum, and RogueMaster is for experienced users who understand compatibility, backups, and rollback. Use official release pages only.

Before installing: confirm hardware revision, charge the device, use a data USB cable, insert a known-good microSD card, back up saved files, and test basic menus after flashing.

9. Bruce-Style Setup Flow

For Bruce-compatible ESP32 products, the process is simple: choose the exact target, flash safely, test basic functions, then learn menus.

Use the Bruce web flasher in Chrome or Edge for supported devices. Select the exact target and follow browser prompts.

M5Burner may be used where a matching Bruce build is available. Choose the correct category and burn only the firmware intended for the connected device.

Advanced local binaries are for users who know the board target and recovery method. If a device does not boot, restore a known-good build for that exact model before trying another image.

Bruce menus may include Wi-Fi, BLE, IR, Sub-GHz/RF, files, scripts, settings, updates, logs, web interface, and device information depending on build.

10. Universal RF Research Tool Manual

This section normalizes the Universal RF Research Tool as a compliance-review and shielded-lab reference. It is not an operating guide for public-space interference.

The main unit is described as a portable aluminum device with antenna bank, power/charging markings, controls, switch bank, cooling area, and status indicators.

Use only as controlled-lab RF education or compliance-review equipment. Do not use it to affect public or private communications, navigation, safety systems, emergency services, or shared networks.

11. Universal RF Research Tool Technical Details

The source manual describes a portable full-band RF research/compliance-review platform using surface-mount components and integrated circuits in an aluminum enclosure with active cooling, power indicator, side controls, antenna/channel switches, and charging socket.

Manual-listed evaluation areas include CDMA/GSM/DCS/PHS, 2G/3G/4G/5G, Wi-Fi 2.4 GHz and 5.8 GHz, GPS/BD-GPS, VHF, and UHF. These are identification references only and do not create permission to transmit, block, or interfere.

The manual lists approximate effective radius claims around 1-15 m, and sometimes 5-20 m depending on conditions. Real results depend on environment, antennas, enclosure, power state, and authorization.

Power notes: built-in battery, roughly 2-3 hours runtime, and roughly 4 hours for a full charge according to the source manual. Use only supplied or approved charging equipment.

Physical notes: host size about 165 x 97 x 45 mm, package size about 240 x 250 x 69 mm, and total kit weight about 1.11 kg.

Heat note: aluminum transfers heat. Stop use if heat becomes unusual, the fan stops, charging is abnormal, or the enclosure is damaged.

12. Universal RF Band References and Safety Screens

Manual-listed bands are identification references only: 700, 800, 900, 1200, 1350, 1500, 1800, 1900, 2000, 2100, 2300, 2400, 2600, 3400, 4700, 4800, and 5800 MHz families appear in the source context.

Before any channel screen, add a compliance gate: approved location, shielded environment, responsible operator, emergency stop, and written test purpose.

Keep channels off by default. Show clear channel status, session timer, log entry, battery/charging status, thermal state, and a visible stop state.

Do not operate near emergency services, hospitals, vehicles, schools, aviation systems, GPS/GNSS receivers, cellular networks, public safety systems, or shared public spaces.

13. Device and Accessory Links

M5Shark Device: <https://m5shark.com/products/m5shark-device-flipper-zero-alternative>

ESP32-DIV V2.0: <https://m5shark.com/products/esp32-div-v2-0>

RF CLOWN V2.0:

<https://m5shark.com/products/finished-rf-clown-v2-0-esp32-wireless-security-tester-kit-3-nrf24l01-ble-bluetooth-analyzer-pen>

nRFBBox-v3: <https://m5shark.com/products/nrfbox-v3-based-on-esp32>

Chameleon Ultra V2:

<https://m5shark.com/products/chameleon-ultra-contactless-smart-card-emulator-rfid-smart-chip-reader-3xcuid-uid-keychain-c>

Mini ESP32 Marauder: <https://m5shark.com/products/esp32-marauder>

ESP32-C5 Marauder:

<https://m5shark.com/products/esp32-c5-marauder-working-with-flipper-zero>

ESP32 Marauder:

<https://m5shark.com/products/esp32-marauder-mar-x-auder-device-wi-fi-bluetooth-offensive-and-defensive-tools-enable-sec>

Universal RF Research Tool:

<https://m5shark.com/products/universal-rf-device-compliance-review>

Universal Signal RF 18-Port:

<https://m5shark.com/products/universal-signal-rf-18-antennas>

NM-RF-Hat for Bruce CYD: <https://m5shark.com/products/nm-rf-hat-for-bruce-cyd>

H4 + R10 SDR Device: <https://m5shark.com/products/h4-r10-two-in-one-sdr-device>

T-Embed CC1101 Plus: <https://m5shark.com/products/t-embed>

TDeck Plus: <https://m5shark.com/products/tsharkplus>

CC1101 Wireless Module:

<https://m5shark.com/products/cc1101-wireless-module-with-sma-antenna-wireless-transceiver-module-for-315-433-868-915>

NRF24L01+PA+LNA Module:

<https://m5shark.com/products/nrf24l01-pa-lna-2-4g-wireless-module-1100-meter-long-distance-with-antenna>

14. Firmware and Technical Reference Links

Bruce website: <https://bruce.computer/>

Bruce web flasher: <https://bruce.computer/flasher>

Bruce firmware GitHub: <https://github.com/BruceDevices/firmware>

Flipper Zero firmware: <https://github.com/flipperdevices/flipperzero-firmware>

Flipper downloads: <https://flipperzero.one/update>

Flipper Lab: <https://lab.flipper.net/>

qFlipper documentation: <https://docs.flipper.net/zero/qflipper>

Firmware update guide: <https://docs.flipper.net/zero/basics/firmware-update>

Unleashed firmware: <https://github.com/DarkFlippers/unleashed-firmware>

Momentum firmware: <https://github.com/Next-Flip/Momentum-Firmware> and
<https://momentum-fw.dev/>

RogueMaster firmware: <https://github.com/RogueMaster/flipperzero-firmware-wPlugins>

Xtreme firmware legacy: <https://github.com/Flipper-XFW/Xtreme-Firmware>

M5Burner: <https://docs.m5stack.com/en/uiflow/m5burner/intro>

ESP32 Marauder: <https://github.com/justcallmekoko/ESP32Marauder>

ESP32-DIV hardware wiki: <https://github.com/cifertech/ESP32-DIV/wiki/Hardware>

nRFBBox: <https://github.com/cifertech/nRFBBox>

RF-Clown releases: <https://github.com/cifertech/RF-Clown/releases>

Chameleon Ultra: <https://github.com/RfidResearchGroup/ChameleonUltra>

CC1101 reference: <https://www.ti.com/product/CC1101>

nRF24L01+ specification:

https://www.sparkfun.com/datasheets/Wireless/Nordic/nRF24L01P_Product_Specification_1_0.pdf

15. Universal Quick Start and Troubleshooting

Quick start: identify the exact device family, charge the device, inspect cables and antennas, use the correct accessory, begin with receive-first or demo modes, log approved tests, then power down and store safely.

Device will not power on: use a known-good 5 V USB source and cable, then inspect the battery area before retrying.

Computer does not detect the device: try a data-capable USB cable, another port, and the correct driver or firmware tool for the model.

Storage not detected: reinsert the card, test a known-good card, and confirm the filesystem is compatible.

Weak radio behavior: check antenna type, connector tightness, distance, obstacles, battery state, and local interference.

Wrong menus or missing features: confirm the firmware build and accessory compatibility for that exact device family.

Device becomes hot: stop testing, power down, disconnect charging, and inspect before reuse.

16. FAQ and Help Checklist

Does this work with Flipper Zero firmware? Only use firmware confirmed for the exact M5Shark / SHARK RF hardware revision. Do not assume every fork or older build will work.

Is Bruce firmware for all devices? No. Bruce is for compatible ESP32 devices and matching board builds. Select the exact target before flashing.

Can I test public networks or signals? No. Use owned devices, lab equipment, or targets where you have clear written permission.

Why did menus change after update? The wrong build or an incompatible version may be installed. Restore a known-good firmware for the exact model and retest basic functions.

Can the Universal RF Research Tool be used anywhere? No. Treat it as compliance-review or shielded-lab equipment only.

Before changing firmware: confirm model/revision, use a data USB cable, charge the battery, insert microSD when needed, check About/Firmware, and verify all RF/RFID/NFC targets are owned or authorized.

17. Quick Reference Notes

Short intro: M5Shark devices are for lawful cybersecurity education, RF learning, hardware development, diagnostics, repair, and authorized lab testing. Choose the exact product family and matching firmware path.

Firmware note: compatibility depends on hardware revision. Check About/Firmware before updating and use only official or project-compatible targets.

Safe use note: work only with devices, tags, remotes, networks, and signals you own or have permission to test. Avoid public spaces, shared systems, emergency environments, or any location where testing is not allowed.

Need help note: prepare exact model, current firmware/About screen, cable and SD-card details, accessory list, and a short issue description.

Universal RF note: use only for authorized shielded-lab RF education and compliance review. Check local laws before purchase or use.

18. Flipper-Style Tools and GitHub Resource Directory

Use these references for lawful firmware management, app discovery, development, repair, and authorized lab testing. Avoid random payload packs or unknown files; prefer official firmware, official apps, and well-known project pages.

Official firmware: github.com/flipperdevices/flipperzero-firmware

Update guide: docs.flipper.net/zero/basics/firmware-update

qFlipper docs: docs.flipper.net/zero/qflipper

qFlipper source: github.com/flipperdevices/qFlipper

Flipper Lab and Apps Catalog: lab.flipper.net and lab.flipper.net/apps

App troubleshooting: docs.flipper.net/zero/apps/troubleshoot

Developer documentation: developer.flipper.net

uFBT build tool: github.com/flipperdevices/flipperzero-ufbt

Official IR database: github.com/flipperdevices/IRDB

Community index: github.com/djsime1/awesome-flipperzero and awesome-flipper.com

19. Flipper-Style Custom Firmware Compatibility Notes

Custom firmware can help advanced learning, but it can also create app mismatch, storage problems, boot issues, or feature gaps if it is not built for the exact device target. Most owners should start with stable/release firmware.

Unleashed firmware: popular custom fork for advanced users. Use only official project releases and confirmed compatible targets.

Momentum firmware: modern custom option. Verify target and install path before flashing.

RogueMaster firmware: bundled apps/plugins for experienced users. Back up files and verify compatibility first.

Xtreme firmware legacy: reference only unless you intentionally need an older build and know how to roll back.

Compatibility checklist: exact model and revision confirmed, battery charged, microSD inserted, data USB cable used, backups complete, firmware/app branch matches the device, and app failures retested on stable release firmware.

20. Flipper Firmware Notice: No Factory Keys Found

The warning "No Factory Keys Found. Secure Enclave is damaged. Some apps will not work." may appear when Flipper-style firmware expects original Flipper factory-key data on compatible M5Shark-style hardware. It usually does not mean the M5Shark is physically broken.

What to do: press OK, Back, or Continue to enter the menu. Do not panic and do not repeatedly install random firmware builds.

Expected behavior: normal M5Shark functions should work when correct firmware, SD card, cable, and accessories are used. Some official Flipper security apps may remain limited.

Best first check: after skipping the warning, open About/Firmware and confirm firmware name, version, device model, microSD status, battery, buttons, and menu operation.

What not to do: do not install unknown key-replacement tools and do not flash random builds. Use only the correct firmware path for the exact hardware revision.

Contact support only if the device cannot boot, buttons fail, storage is not detected, or the warning prevents normal menu access.

21. Safety and Legal Notice

This guide is for lawful education, development, repair, interoperability study, and authorized testing. It is not permission to test systems, signals, cards, remotes, locks, computers, or networks without authorization.

You are responsible for local laws, RF rules, access-control rules, privacy rules, import restrictions, battery safety, and permission to test. If you are not sure a test is allowed, do not run it.

Simple rule: own it, have permission, use the right antenna or accessory, stay in the lab, and stop if behavior is unexpected.

Universal RF rule: use only as a controlled, authorized lab safety reference. Do not affect public/private communications or safety systems.

Final checklist: identify exact device family, use only authorized targets, begin with receive-first learning, back up files before updates, and stop immediately if charging or heat behavior is abnormal.